

EXPERIMENTS ON ESTIMATION IN GAMMA POINT PROCESSES WITH EXPONENTIAL COVARIANCE

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1. Introduction

In a gamma point process whose distances have covariance function $R(r) = \sigma^2 \rho^{|r|}$, an estimator of ρ is C_1/C_0 , where C_r is an estimator of $R(r)$. (Linhart (1970).) It occurred to us that it might be of advantage to also use covariances of lag larger than 1 to form an estimator of ρ , in particular if $R(r)$ is not quite, but almost, of the exponential type given above. The natural choice would be to fit, by the method of least squares, a function ρ^r to the observed values of C_r/C_0 , for $r=1,2,\dots,k$, where k is suitably chosen. Any analytical study of the performance of such an estimator (under the hypothesis that the covariance is exponential, and even more so under the hypothesis that the covariance function is only approximately exponential) is extremely difficult. We undertook therefore to clarify this matter by a few Monte Carlo experiments. This provided also an opportunity to check experimentally on some of the theoretically obtained results (Linhart (1970)): on the approximate standard errors, and on the influence of using lengths instead of distances for the various estimators. This note is a summary of the results of these experiments.

2. Methods used to generate gamma point processes

If the vectors $Z_i = (Z_{ij})$ ($i=1,2,\dots,p$; $j=1,2,\dots,n$) are independently and identically distributed as $N(0, \Sigma_n)$, where $\Sigma_n = (\sigma_{ij})$ is positive definite and $\sigma_{ij} = S(|i-j|)$, then $X = (X_j)$, where $X_j = \sum_{i=1}^p Z_{ij}^2$, has a multivariate gamma distribution with characteristic function $|\Gamma(2iD(u_1, u_2, \dots, u_n) \Sigma_n)^{-1} p/2|$ (Krishnaiah and Rao (1961)). Realisations of X can then be regarded as realisations of size n of the process of distances in a gamma point process with covariance function $R(r) = pS^2(|r|)$.

The vectors Z_i were generated using the moving average representation $Z_{ij} = \sum_{k=0}^{\infty} a_k Y_{ij-k}$, where the Y_{ij} are $NID(0, \sigma_y^2)$; Z_i is then $N(0, \Sigma_n)$ with $\sigma_{ij} = S(|i-j|) = \sigma_y^2 \sum_{k=0}^{\infty} a_k a_{k+|i-j|}$ (Cox and Miller (1965)). In the special case of exponential covariance, $S(|i-j|) = S(0)\rho^{|i-j|/2}$, the moving average series is infinite with $a_k = \rho^{k/2}$, $k=0,1,2,\dots$, and it is easier to use the autoregressive representation, which in this case is very simple: $Z_{ij} = \rho^{1/2} Z_{ij-1} + Y_{ij}$, where $\sigma_y^2 = S(0)(1-\rho)$. To generate vectors with covariance functions which are only approximately exponential we used the moving average representation with some decreasing sequence for a_k for $k=0,1,\dots,K$ and $a_k = 0$ for $k>K$. The NID variables Y_{ij} were generated according to Knuth (1969, p. 104).

In order to generate lengths we chose a constant T (order of magnitude about 100 $E(X_j)$) and used as lengths $L_k \equiv X_{s+k}$, $k=1,2,\dots,n$, where s is the smallest integer such that $\sum_{j=1}^{s+1} X_j \geq T$.

3. An alternative method to estimate rho

As mentioned in the introduction, instead of using $\hat{\rho} = C_1/C_0$ one could use the estimator $\hat{\rho}(k)$, the value of ρ which minimises

$$\sum_{r=1}^k (\rho^r - C_r/C_0)^2$$

Obviously $\hat{\rho} \equiv \hat{\rho}(1)$. It turned out that the best numerical method to find $\hat{\rho}(k)$ from a given set of values of C_r/C_0 , $r = 1, 2, \dots, k$, is a simple grid search.

We generated about one hundred realisations of n distances each of a gamma point process with exponential covariance and calculated $\hat{\rho}(k)$ and $\Delta(k) = \rho - \hat{\rho}(k)$ for $k = 1(1)20$. The one hundred mentioned processes had parameters in the ranges: $\rho = .2(.1).9$; $P = 5, 20, 50$; $\sigma_y^2 = 1, 5$; $n = 100, 250, 500$. Generally there was very little variation in $\Delta(k)$ with varying k , and for k larger than about ten $\Delta(k)$ remained constant (though this depended on ρ). Sometimes $\Delta(1)$ was slightly larger than the other differences, sometimes slightly smaller. On the whole there was very little difference in performance between the estimators, they seem to have about the same standard error, and the standard error is very small, even for $n = 100$. In the range of parameters which we used, $k = 10$ seems to be a convenient value to choose.

We also generated distances of gamma processes whose covariance function was only approximately exponential. We made about twenty trials with $n = 100, 200, 500$, ρ between .4 and .9, $P = 10, 20$ and $\sigma_y^2 = 1$. In each case we calculated the covariance function of the process of distances (using the coefficients a_k and the formula given in Section 2) and fitted by least squares a "best" value of ρ , we denote it by ρ_0 . We then generated n distances from this process, estimated the covariance function, by C_r , calculated $\hat{\rho}$ and $\hat{\rho}(K)$, and compared these two estimators with ρ_0 . For small values of n (100 or less) there was little or no difference in the performance of the two estimators: the standard errors were such that any differences could not be detected. For large values of n (200 or more) the differences were marked: $\hat{\rho}(K)$ was always found to be nearer to ρ_0 . Three typical cases are summarised below.

r	0	1	2	3	4	5	6	
a_r	.66	.44	.39	.30	.29	.22	.15	$n = 100, p = 20, \sigma_y^2 = 1$
$R(r)/R(0)$	1.00	.55	.36	.20	.11	.04	.01	$\hat{\rho} = .61$
ρ_0	1.00	<u>.57</u>	.32	.19	.11	.06	.03	$\hat{\rho}(6) = .54$
a_r	.68	.45	.41	.35	.28	.21	.15	$n = 200, p = 20, \sigma_y^2 = 1$
$R(r)/R(0)$	1.00	.56	.37	.21	.10	.04	.01	$\hat{\rho} = .46$
ρ_0	1.00	<u>.58</u>	.34	.20	.11	.07	.04	$\hat{\rho}(6) = .54$
a_r	.43	.36	.31	.26	.21	.15	.10	$n = 500, p = 20, \sigma_y^2 = 1$
$R(r)/R(0)$	1.00	.65	.40	.22	.10	.03	.01	$\hat{\rho} = .56$
ρ_0	1.00	<u>.61</u>	.37	.23	.14	.09	.05	$\hat{\rho}(6) = .59$

The results of the experiments suggest that no damage can be done if $\hat{\rho}(k)$ is used ($k > 1$; convenient: $k = 10$) instead of $\hat{\rho}$, even if the covariance is exponential. If the covariance function is only approximately exponential $\hat{\rho}$ can be very misleading, whereas $\hat{\rho}(k)$ performs satisfactorily. It seems, therefore, advisable to use, in any case, $\hat{\rho}(k)$ rather than $\hat{\rho}$ as estimator.

4. Validity of standard error formulae

In Linhart (1970, Section 5.2) approximate standard errors are given for the estimators $\hat{\mu}$, $\hat{\sigma}^2$, $\hat{\rho}$. In order to check the validity of these formulae we generated about twenty realisations each for ten combinations of the parameters n, ρ, p, σ_y^2 . We estimated the standard errors from the twenty estimators each and compared them with the calculated standard errors. The standard error formulae for $\hat{\mu}$ and for $\hat{\sigma}^2$ seem to be very good. The formula for $\hat{\rho}$ seems to give slightly too large values. The average values – taken over the ten available cases – of the ratios (calculated standard errors)/(estimated standard errors) were 1.02, 1.03 and 1.17 respectively. We also generated one realisation each for 162 processes, where the parameters ranged over the values given in Section 3, and calculated the ratios (parameter-estimator)/(calculated standard error). These ratios should be approximately normally distributed with zero mean and standard deviation one. Below are the means and standard deviations of the mentioned ratios, taken over all available results.

	Mean	Standard Deviation	Number of cases
$\hat{\mu}$	0.06	0.94	162
$\hat{\sigma}^2$	-0.03	0.97	162
$\hat{\rho}$	0.01	0.85	162

This confirmed the previously obtained impression that the standard error formulae for $\hat{\mu}$ and for $\hat{\sigma}^2$ are very good and that the calculated standard error of $\hat{\rho}$ tends to be too large.

5. The bias introduced by using lengths

We calculated for a few cases the bias of $\bar{L} = \sum_{i=1}^n L_i/n$ and an approximation to the bias of $C_0 = \sum_{i=1}^n (L_i - \bar{L})^2/n$, where L_i are lengths.

The expectation of $\bar{L}$ is given in Linhart (1970, Section 5.3). The following result can also be verified:

$$\begin{aligned}
 E(C_0) &= E\left(\sum_{i=1}^n L_i - E(L_i)\right) (L_{i+r} - E(L_{i+r})) / (n-r) \\
 &= \sigma^2 \rho^r \left(1 + \frac{1}{n-r} \frac{\sigma^2}{\mu^2} \frac{\rho(1-\rho^{n-r})(2+\rho+\rho^{n-r+1})}{(1-\rho^2)}\right)
 \end{aligned}$$

The table below gives (1): $100 (E(\bar{L}) - E(X_i))/E(X_i)$
 and (2): $100 (E(C'_0) - \text{Var}(X_i))/\text{Var}(X_i)$
 for some values of ρ and n .

n	5		10		50	
ρ	(1)	(2)	(1)	(2)	(1)	(2)
.1	.22	.42	.11	.21	.02	.04
.3	2.56	4.53	1.28	2.27	.25	.45
.7	27.17	41.28	11.74	12.50	3.26	5.19

We also generated lengths and distances from processes with identical parameters and compared estimators of μ , σ^2 and ρ based on lengths with those based on distances. For our values of n (50, 100, 250) the bias could not be noticed since it was much smaller than the standard error.

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